# MAYA MAES-JOHNSON

Baton Rouge , Louisiana, USA
+1 (719) 744-4192 ⋄ mmaesj1@lsu.edu

## EDUCATION

**Louisiana State University (LSU)** August 2025 - Expected May 2030
*PhD Student in Oceanography & Coastal Sciences* *Overall GPA: 4.0 / 4.0*
*Dissertation: River delta network structure & dynamics*

**Colorado School of Mines (CSM)** December 2024
*BS in Applied Mathematics* *Overall GPA: 3.4 / 4.0*
*Minor in Public Affairs*

## TECHNICAL SKILLS

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|---|---|
| **Computer Languages** | MATLAB, Python, R, UNIX |
| **Modeling Software & Tools** | ArcGIS, WRF, TopoTide |
| **Field & Laboratory Skills** | ADCP hydrographic surveys, stratigraphic measurement, sediment filtration |

## PROFESSIONAL EXPERIENCE

**Coastal Hydrology, Hydrodynamics, and Oceanography Lab, LSU** August 2025 - Present
*Graduate Research Assistant; Advised by Dr.Matthew Hiatt*

- Studying the evolution of channel network structure and water transport time using hydrodynamic modeling and particle tracking
- Developing methodology to quantitatively analyze the structure and flux behavior of river deltas

**Coastal Ecosystem Design Studio (CEDS) Internship, LSU** Summer 2026
*Summer Intern; Advised by Dr. Traci Birch*

- Worked with a team to research and develop restoration design concepts of river siphons in Barataria Bay and strategic vegetation plantings in Neptune Pass
- Conducted field interviews with residents of Plaquemines Parish to integrate community input to design concepts
- Gained experience using Adobe Creative Suite to create visuals for final poster, paper, and presentation

**Sedimentology Research Lab, CSM** Jun 2023 - December 2024
*Undergrad Research Assistant; Advised by Dr. Piret Plink-Björkland, Molly O'Halloran (PhD student)*

- Participated in five weeks of fieldwork in Utah to collect stratigraphic data from the early Paleogene rock record
- Extracted profiles of modern rivers exhibiting Froude supercritical flow to study relationship between inertia-driven flows and slope

**Jackson School of Geosciences, UT Austin** Summer 2024
*Research Trainee; Advised by Dr. John Goff, Carson Miller (PhD Student)*

- Analyzed and visualized eight years of bathymetric data to observe changes in subaqueous estuarine dunes in Port Aransas, Texas
- Participated in weekly research trainee cohort workshops for professional development

**MoWaTer Data Science Fellowship, Baylor University** Summer 2022
*Data Science Fellow; Advised by Dr. Amanda Hering (Baylor), Dr. Doug Nychka (CSM)*

- Collaborated on a team of three students to analyze time-series data and create future predictions of carbon-to-sulfate mass ratio at wastewater treatment plants in Colorado
- Interpreted statistical results and created visualizations to present to stakeholders

## HONORS, AWARDS, AND SCHOLARSHIPS

- CSM Undergraduate Research Travel Grant (2024)
- The Hydrographic Society of America (THSOA) Houston Chapter Scholarship (2024)
- Fall 2024 Undergraduate Research Symposium 3rd Place (2024)
- Mines Undergraduate Research Fellow (2024)
- Fall 2023 Undergraduate Research Symposium People's Choice Award (2023)
- Summer Undergraduate Research Fellow (2023)
- McBride Outstanding Sophomore Award (2022)
- McBride Honors Program (2021-2024)
- Thorson First-Year Honors Program (2020-2021)
- Hurliman Foundation Scholarship (2020)

## RELEVANT COURSES

Deltaic Geology
Geological Oceanography
Gravity Waves in Shallow Water
Chemical Oceanography
Physics of Sediment Transport
Wetland Hydrology
Physical Oceanography
Modeling the Marine Atmosphere
Scientific Computing

## PEER-REVIEWED PUBLICATIONS

1. Strong, S., **Maes-Johnson, M.** (2024). "Nonlinear Dynamics of Mountain Pine Beetle Populations: Discussion of Forestry Policy, a Survey of Existing Mathematical Models, and Code Base Demonstration," CODEE Journal: Vol. 17, Article 8.

## CONFERENCE PRESENTATIONS

1. **M. Maes-Johnson**, M. O'Halloran, P. Plink-Björklund (2024), American Geophysical Union (AGU) Annual Conference. "Froude Supercritical Flow Occurrences in Modern Mobile Bed River Systems" (Poster)
2. **M. Maes-Johnson**, C. Miller, M. Davis, D. Duncan, J. Cisneros, J. Goff, S. Gulick, C. Lowery (2024), American Geophysical Union (AGU) Annual Conference. "Impacts of Dredging and Storm Events on Subaqueous Estuarine Dunes in Port Aransas, Texas" (ePoster)

## OUTREACH

**Environmentors (LSU)** August 2025 - April 2026
*Mentored a high school student on a project focused on PFAS contamination in water bodies across Baton Rouge*

**Girls Lead the Way (CSM)** January 2024
*Instructed a mini-lesson on field measurement in stratigraphy and assisted with flume demonstration during SWE's Girls Lead the Way conference*

**Nine Mile Canyon Stewardship Day** September 2024
*Presented on geological history and current research in Nine Mile Canyon, Utah to over 100 Stewardship Day attendees*

## TRAINING AND WORKSHOPS

CSDMS Earth Surfaces Processes Institute (ESPIn), University of Colorado August 2026